

XIANG LI

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EDUCATION

Peking University, Beijing, China

Sep. 2018 – Jul. 2023

Ph.D. in Statistics, School of Mathematical Sciences

Advisor: Prof. Zhihua Zhang

Peking University, Beijing, China

Sep. 2014 – Jul. 2018

B.S. in Statistics, School of Mathematical Sciences

B.A. in Economics, National School of Development

EXPERIENCE

University of Pennsylvania, Philadelphia, USA

Aug. 2023 – Present

Postdoctoral Researcher, Department of Statistics and Data Science

Working with Prof. Weijie Su and Prof. Qi Long

University of Toronto, Toronto, Canada

Feb. 2022 – Feb. 2023

Visiting Ph.D. Student, Department of Statistical Sciences

Working with Prof. Qiang Sun

Face++ (Megvii), Beijing, China

Jun. 2018 – Nov. 2018

Research Intern, Algorithm Group

Mentor: Dr. Shuchang Zhou

ToSimple, Beijing, China

Mar. 2018 – May. 2018

Research Intern, Algorithm Group

Mentor: Dr. Naiyan Wang

SELECTED AWARDS

- IMS New Researcher Travel Award *2025*
- Outstanding PhD Graduates, Peking University *2023*
- President Scholarship for Excellent Ph.D Student, Peking University *2018, 2020-2021*
- Outstanding reviewer in NeurIPS, top 8% *2021*
- National Scholarship, China *2017, 2019*
- Travel Award *NeurIPS 2019, AAAI 2020*
- Outstanding Graduates, School of Mathematical Sciences, Peking University *2018*
- First prize in National Undergraduate Mathematical Modeling Contest, China *2016*

RESEARCH INTERESTS

My research lies at the intersection of statistics, optimization, and machine learning. Currently, I focus on principled and scalable methodologies for generative AI—particularly large language models—with the goal of advancing theoretical understanding and developing insights that support safe, robust, and reliable deployment. Specifically, my research spans the following directions:

- **Statistical Foundations of Generative AI**

- **Provenance and Safety Mechanisms** Designing watermarking schemes and developing detection rules for tracing and verifying AI outputs, with the broader aim of building scalable safety mechanisms that safeguard provenance, accountability, and responsible deployment.
- **Fundamental Principles** Modeling generative AI systems as black boxes and developing statistical tools to analyze input–output relationships, focusing on core issues such as evaluation, uncertainty quantification, and model collapse.
- **Statistical Methods for Reliable Learning**
 - **Robustness to Adverse Conditions** Developing efficient algorithms resilient to nonstandard conditions such as heavy-tailed noise, corruption, and data heterogeneity.
 - **Uncertainty Quantification in Adaptive Environments** Designing statistical methods for streaming data and adaptive decision-making, ensuring valid statistical inference in dynamic environments.
- **Optimization for Federated Learning** Designing communication-efficient algorithms and analyzing convergence behavior under heterogeneous data.

PUBLICATIONS

* indicates equal contribution; ** indicates alphabetical order.

Journal papers

- [J1] **A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules** [\[pdf\]](#) [\[arxiv\]](#)
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
Annals of Statistics, 53 (1): 322-351, Feb. 2025
- [J2] **Robust Detection of Watermarks for Large Language Models Under Human Edits** [\[arxiv\]](#)
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
To appear in *Journal of the Royal Statistical Society Series B*
- [J3] **Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts** [\[arxiv\]](#)
Xiang Li, Garrett Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su
Major revision at *Journal of the American Statistical Association*.
- [J4] **Debiasing Watermarks for Large Language Models via Maximal Coupling** [\[pdf\]](#) [\[arxiv\]](#)
Yangxinyu Xie, Xiang Li, Tanwi Mallick, Weijie J. Su, Ruixun Zhang
Journal of the American Statistical Association, 1–11, 2025
- [J5] **Convergence and Inference of Stream SGD, with Applications to Queueing Systems and Inventory Control** [\[arxiv\]](#)
Xiang Li*, Jiadong Liang*, Xinyun Chen, Zhihua Zhang
Minor revision at *Operation Research*
- [J6] **A Random Projection Approach to Personalized Federated Learning: Enhancing Communication Efficiency, Robustness, and Fairness** [\[pdf\]](#)
Yuze Han**, Xiang Li**, Shiyun Lin**, Zhihua Zhang**
Journal of Machine Learning Research, 25 (380): 1-88, Dec. 2024
- [J7] **Variance-aware Decision Making with Linear Function Approximation with Heavy-tailed Rewards** [\[pdf\]](#) [\[arxiv\]](#)
Xiang Li, Qiang Sun
Transactions on Machine Learning Research, Apr. 2024

- [J8] **FedPower: Privacy-Preserving Distributed Eigenspace Estimation** [\[pdf\]](#) [\[arxiv\]](#)
Xiao Guo*, **Xiang Li***, Xiangyu Chang, Shusen Wang, Zhihua Zhang
Machine Learning, 113 (11): 8427-8458, Sep. 2024

Conference papers

- [C1] **On the Empirical Power of Goodness-of-Fit Tests in Watermark Detection**
Weiqing He* and **Xiang Li***, Tianqi Shang, Li Shen, Weijie J. Su, Qi Long
Neural Information Processing Systems (NeurIPS) 2025, Spotlight
- [C2] **Mitigating the Privacy–Utility Trade-off in Decentralized Federated Learning via f -Differential Privacy**
Xiang Li, Chendi Wang, Buxin Su, Qi Long, Weijie J. Su
Neural Information Processing Systems (NeurIPS) 2025, Spotlight
- [C3] **A Statistical Analysis of Polyak-Ruppert-Averaged Q-Learning** [\[pdf\]](#)
Xiang Li, Wenhao Yang, Jiadong Liang, Zhihua Zhang, Michael I. Jordan
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023
- [C4] **Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards** [\[pdf\]](#)
Qingyuan Yu, Euijin Baek, **Xiang Li**, Qiang Sun
Conference on Uncertainty in Artificial Intelligence (UAI) 2025
- [C5] **Statistical Analysis of Karcher Means for Random Restricted PSD Matrices** [\[pdf\]](#)
Hengchao Chen, **Xiang Li**, Qiang Sun
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023
- [C6] **Statistical Estimation and Online Inference via Local SGD** [\[pdf\]](#)
Xiang Li, Jiadong Liang, Xiangyu Chang, Zhihua Zhang
Conference on Learning Theory (COLT) 2022
- [C7] **Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective** [\[pdf\]](#)
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022, Spotlights
- [C8] **Personalized Federated Learning towards Communication Efficiency, Robustness and Fairness** [\[pdf\]](#)
Shiyun Lin*, Yuze Han*, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022
- [C9] **Communication-Efficient Distributed SVD via Local Power Iterations** [\[pdf\]](#)
Xiang Li, Shusen Wang, Kun Chen, Zhihua Zhang
International Conference on Machine Learning (ICML) 2021
- [C10] **Finding the Near Optimal Policy via Reductive Regularization in MDPs** [\[pdf\]](#)
Wenhao Yang, **Xiang Li**, Guangzeng Xie, Zhihua Zhang
Workshop on Reinforcement Learning Theory, ICML 2021
- [C11] **On the Convergence of FedAvg on Non-IID Data** [\[pdf\]](#)
Xiang Li*, Kaixuan Huang*, Wenhao Yang*, Shusen Wang, Zhihua Zhang
International Conference on Learning Representations (ICLR) 2020, Oral presentation
- [C12] **Do Subsampled Newton Methods Work for High-Dimensional Data?** [\[pdf\]](#)
Xiang Li, Shusen Wang, Zhihua Zhang
AAAI Conference on Artificial Intelligence (AAAI) 2020

- [C13] **A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning** [\[pdf\]](#)
Wenhao Yang*, **Xiang Li***, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2019

Submitted

- [S1] **Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?** [\[arxiv\]](#)
Xiang Li, Jiayi Xin, Qi Long, Weijie J. Su
- [S2] **Uncertainty Quantification of Data Shapley via Statistical Inference** [\[pdf\]](#)
Mengmeng Wu, Zhihong Liu, **Xiang Li**, Ruoxi Jia, Xiangyu Chang
- [S3] **Finite-Time Decoupled Convergence in Nonlinear Two-Time-Scale Stochastic Approximation** [\[arxiv\]](#)
Yuze Han**, **Xiang Li****, Zhihua Zhang**
- [S4] **Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation** [\[arxiv\]](#)
Yuze Han, **Xiang Li**, Jiadong Liang, Zhihua Zhang
- [S5] **Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data** [\[arxiv\]](#)
Xiang Li, Jiadong Liang, Zhihua Zhang
- [S6] **Privacy-Preserving Community Detection for Locally Distributed Multiple Networks** [\[arxiv\]](#)
Xiao Guo, **Xiang Li**, Xiangyu Chang, Shujie Ma

Technical reports

- [T1] **Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: A Jump Diffusion Approach** [\[arxiv\]](#)
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
- [T2] **Communication-Efficient Local Decentralized SGD Methods** [\[arxiv\]](#)
Xiang Li, Wenhao Yang, Shusen Wang, Zhihua Zhang

PRESENTATIONS

- *Robust Watermark Detection and Efficient Proportion Estimation for Human-Edited Watermarked Text*
 - Jun. 2025, ICSA Applied Statistics Symposium, University of Connecticut.
 - Aug. 2025, Joint Statistical Meetings, Nashville, Tennessee.
- *Optimal Robust Detection for Gumbel-Max Watermarks Under Contamination*
 - Jul. 2024, IMS-China International Conference on Statistics and Probability, Yinchuan.
 - Sep. 2024, Penn conference on Big Data in Biomedical & Population Health Sciences.
 - Sep. 2024, Warren & ASSET Center Research Mixer, UPenn.
- *A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules*
 - May 2024, the NUS IMS workshop on statistical machine learning for high dimensional data, Singapore.
 - Jul. 2024, the 2nd joint conference on statistics and data science in China (JCSDS), Kunming.

- Nov. 2024, the conference on statistical learning and data science (SLDS), Newport Beach, California.
- *Complete Asymptotic Analysis for Projected Stochastic Approximation and Debiased Variants*
 - Sep. 2023, the Allerton conference at the University of Illinois, Allerton Park & Retreat Center.
- *Polyak-Ruppert-Averaged Q-Learning is Statistically Efficient*
 - Jul. 2022, the RL workshop in Shanghai University of Finance and Economics.
- *Statistical Estimation and Inference via Local SGD in Federated Learning*
 - Nov. 2021, the 14-th China-R conference.
- *A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning*
 - Jun. 2019, the machine learning workshop at Peking University.

ACADEMIC SERVICE

- Conference review: NeurIPS, ICML, ICLR, UAI, AISTATS, IJCAI.
- Journal review: Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), Journal of the American Statistical Association (JASA), Annals of Applied Probability (AOAP), Operational Research (OR), IEEE Transactions on Automatic Control, IEEE Open Journal of Signal Processing, IEEE Journal on Selected Areas in Communications, Transactions on Parallel and Distributed Systems, World Wide Web (WWW).
- Event organization:
 - Aug. 2024, Session Chair for “Theoretical and Algorithmic Developments in Federated Learning” at the Modeling and Optimization: Theory and Applications (MOPTA) conference.

TEACHING EXPERIENCES

- Lecturer, Short course at the 2025 ICSA Applied Statistics Symposium, Summer 2025 [\[slides\]](#)
- Teaching Assistant, *Linear Algebra*, Spring 2021
- Teaching Assistant, *Reinforcement Learning: Theory and Algorithms*, Fall 2019